

Name: _____ Neptun code: _____

(KTXFI2EBNF) Physics I. Exam 1
May 26, 2026

I. Mechanics

1. What does Newton's first law state? What does Newton's second law describe? What is Newton's third law? (5 points)
2. What are conservative forces? What are dissipative forces? (5 points)
3. What is momentum? What is conservation of momentum? (5 points)
4. What is escape velocity? (5 points)

II. Thermodynamics

5. What does the first law of thermodynamics state? What is the second law of thermodynamics? (5 points)
6. What is thermal expansion? What is linear expansion? What is volumetric expansion? (5 points)
7. What is the ideal gas model? What is the ideal gas equation conceptually? (5 points)
8. What is the Carnot cycle? (5 points)

III. Electrostatics, Waves, and Optics

9. What is Coulomb's law? (5 points)
10. What is the Doppler effect? (5 points)

Grades: 0–25 points: Fail (1), 26–30 points: Pass (2), 31–37 points: Satisfactory (3), 38–45 points: Good (4), 46–50 points: Excellent (5).

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